

Run a length of RJ-45 wire from the port on your modem to the network card socket on your motherboard.

Your NIC will have a device node `eth0`. Add the following two lines to the `/etc/resolv.conf` file:

```
nameserver 208.67.222.222
nameserver 208.67.220.220
```

Execute the following command:

```
ifconfig eth0 192.168.1.2
route add default gw 192.168.1.1 ## assuming this is your
gateway address—the IP address of your router
ifconfig eth0 up
```

Now reboot the system and try to browse the Internet. Everything should work.

At this stage, run an `apt-get update` and `apt-get dist-upgrade` to upgrade all installed packages to their latest versions.

Stage 4: Let the world welcome you

Yeah, you got it—DNS.

You can have your own DNS if you want, and I'll cover that in the last part of the series. But right now, we will not have our own DNS server. What we are going to do is sign up for a DNS service provider.

First, register a domain. You can get one for free at www.co.cc. Sign up for a free account. You can have up to two domains per account. They are in the form `yourdomain.co.cc`. Mine is www.theossfreak.co.cc, but don't freak out here. Not all domains are free, some require up to \$900 per year. The trick is to make a simple spelling mistake. MangoesAreSticky comes for \$7 a year, but take the `e` out of Mangoes and the domain is free for life.

When you set up your domain, use the first of three configuration options. Set up the nameservers as `ns1.everydns.com` through `ns4.everydns.com` (4 in all, you need to add 2).

Now go to www.everydns.com. Sign up for another free account. You will be signed in immediately. Add your domain as a basic domain (on the left). Give the full name without the `www`, like `yourdomain.co.cc`. The right pane will now show a table with a single 'A' (the IP address) and a 'CNAME' record. You need to add one more first and then remove the other two.

Remember the static IP address that your ISP gave you when you asked for it? Just below the table is the form for adding a record. Give the domain name without the `www` in the first field. Now let the record type be A. In the value field, give your static IP address. If it's BSNL, then it most probably starts with 59. Then commit the changes. Your existing A record will change to reflect the new settings with a TTL or Time To Live of 3600 seconds, or 60 minutes. We are done.

What now?

Because server connectivity is brilliantly tough, you are bound to hit problems. You can try out Internet forums, try your own luck, or you can e-mail me. The problems vary greatly and are dependent on many things, such as the USB drive not being found, or you not being able to connect to the Internet. Take these problems head-on and overcome them. Remember that this server has no enterprise support.

We are not building this server as an expensive Internet browsing machine. We will eventually set up the server role-by-role. I did want to set up every piece of software from the sources, but ran out of patience. Hundreds of dependencies, and equally high compile times, made it an exhausting task. Thus the controversial choice of the OS. (Whod use a desktop OS as a server?)

There are advantages to using Ubuntu. It's a very simple to use Debian-based distribution. We can tweak it exactly to our needs. Even now if you think KDE4 is too heavy for a server, issue `apt-get install xubuntu-desktop` to go with XFCE 4.4.2. Brilliant!

Next month, we will start setting up our server. We will first set up the DHCP services. Then comes the setting up of a TFTP server as the first real server role. (TFTP is kind of useless, but it is used to remote-boot PCs over the network using PXE.) We will also set up a firewall, following which we will set up a basic LLMPPP (Linux-LightTPD-MySQL-PHP-Python3-Perl) stack for our Web apps. (If you doubt LightTPD's power, it is used by both Wikipedia and YouTube, and they're seldom, if ever, down!)

In the subsequent months, we will start by setting up a complete Web and database server, an e-mail server, fax, IRC and Jabber (for chat), virtualisation, file storage (FTP), print and media streaming, in that order. You can skip any function you like, and you can ask me to add roles that you may want. I'm open to suggestions and contributions, in the true spirit of FOSS.

For now, go and purchase another network card, and a network hub. Configure the new NIC as `eth1` with an IP address of 192.168.1/24. This will be the IP address that all the PCs in your network will know as the server. Connect `eth1` to the hub. All the PCs in your network will connect to the hub in a Star-Bus Topology [see sidebar].

I also suggest that you do some research on basic networking, the Ring and Star topologies, and CSA (Client Server Architecture). This will help, as you will know what you are doing.

See you next month! 

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The author is a 14-year-old student studying in Class 8. He is a logician (as opposed to magician), a great supporter of Free Software and loves hacking Linux. Other than that, he is an experienced programmer in BASIC and can also program in C++, Python and Assembly (NASM Syntax).